Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AI7DE41]

SEVENTHSEMESTER B.Tech EXAMINATION, DECEMBER-2021

ADVANCED NATURAL LANGUAGE PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain in detail about the seq2seq Model with neat diagram	**10**
	b.	List the advantages and disadvantages of deep learning model and the classical supervised learning model.	**04**
2.	**a.**	Consider the following sentences. **i.** <s>I am Ram</s> **ii.** <s>Ram is busy</s> **iii.** <s>I like coffee very much.</s> Calculate the probability using Bigram model for the following sentence. Ram likes coffee.	**12**
	b.	What is one-hot encoding. Explain with a simple example	**02**
3.	**a.**	List and Explain various types of PoS tagging methods.	**10**
	b.	What is closed and open class categories in PoS	**04**
		OR	
4.	**a.**	Define and describe NER in NLP	**10**
	b.	Define word analogy. Give a parallelogram representation for the words 'man', 'woman', 'king' and 'queen'.	**04**
5.	**a.**	What is attention mechanism in seq2seq model? List and explain various attention mechanism.	**08**
	b.	Explain Skip-Gram model processing steps with neat sketch	**06**
		OR	
6.	**a.**	Construct the corpus for the following sentence using Skip-Gram model. "No Chatbot without NLP Applications"	**10**
	b.	What is Glove? Construct the linear structure graph for the following word pairs. (man, woman), (son, daughter), (sir, madam)	**04**
7.	**a.**	What is Deep sequence Model? List and explain various types of Deep sequence representation with neat diagram?	**10**
	b.	Explain the role of Dynamic Programming in Machine Learning.	**04**
		OR	
8.	**a.**	Explain about BiLSTM-CRF Model with Neat diagram?	**10**
	b.	Define Chatbot. List and explain the types of chatbots.	**04**